DO NOW — RECALL

# Lesson 095 · CPU Pipelining: Overlap and Hazards

Name: _____ Class: _____ Date: _____

**1. What is pipeline throughput?**

**2. What is a hazard?**

**3. Which CPU feature may reduce RAM accesses?**

**4. A D flip-flop has D=0 and Q=1 just before the active rising edge. What is Q after?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Rate of completed instructions.

**2. Model answer**

Dependency/resource/control issue that interrupts pipeline.

**3. Model answer**

Cache.

**4. Model answer**

Q becomes0, sampled at the active edge.

**By the end, you can…**

• Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

• With four independent instructions and three one-cycle stages, ideal completion is 3+4−1=6 cycles; serial non-overlap takes 4×3=12.

• Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

**Vocabulary — know these terms**

pipeline — Stages overlapping different instructions.

throughput — Completed work per time.

data hazard — Dependency on unavailable result.

control hazard — Branch uncertainty interrupting fetch.

NEW LEARNING

# Read, annotate and explain

### Observe

Pipelining splits instruction processing into stages; several instructions can occupy different stages during the same clock cycle.

### Explain

With four independent instructions and three one-cycle stages, ideal completion is $3+4-1=6$ cycles; serial non-overlap takes $4\times3=12$.

### Evaluate

Data dependencies, branching and resource conflicts can stall or flush a pipeline. Throughput benefits need ideal assumptions, not automatic latency reductions.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning

**CPU Pipelining: Overlap and Hazards · STAGE 1**

| | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| I1 | F | D | E | | | |
| I2 | | F | D | E | | |
| I3 | | | F | D | E | |
| I4 | | | | F | D | E |

**Four independent instructions complete after 6 ideal cycles.**

CPU Pipelining: Overlap and Hazards · STAGE 1; ideal pipeline stages for four independent instructions

### New Learning

**CPU Pipelining: Overlap and Hazards · STAGE 2**

| | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| I1 | F | D | E | | | |
| I2 | | F | D | E | | |
| I3 | | | F | D | E | |
| I4 | | | | F | D | E |

**Four independent instructions complete after 6 ideal cycles.**

CPU Pipelining: Overlap and Hazards · STAGE 2; ideal pipeline stages for four independent instructions



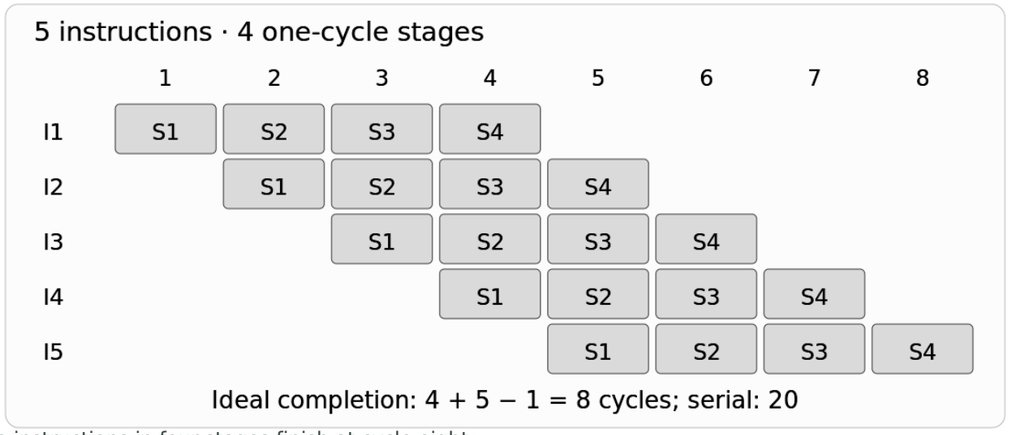


# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Five independent instructions use a four-stage ideal pipeline. Calculate completion.**

4+5−1=8 cycles, compared with 4×5=20 serial stage cycles.




Five instructions in four stages finish at cycle eight

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Explain the mechanism
4+5−1=8 cycles, compared with 4×5=20 serial stage cycles.
Assumes equal one-cycle stages, no resource conflicts or stalls.





# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning

**CPU Pipelining: Overlap and Hazards · STAGE 3**

| | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| I1 | F | D | E | | | |
| I2 | | F | D | E | | |
| I3 | | | F | D | E | |
| I4 | | | | F | D | E |

**Four independent instructions complete after 6 ideal cycles.**

CPU Pipelining: Overlap and Hazards · STAGE 3; ideal pipeline stages for four independent instructions

### New Learning

**CPU Pipelining: Overlap and Hazards · STAGE 4**

| | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| I1 | F | D | E | | | |
| I2 | | F | D | E | | |
| I3 | | | F | D | E | |
| I4 | | | | F | D | E |

**Four independent instructions complete after 6 ideal cycles.**

CPU Pipelining: Overlap and Hazards · STAGE 4; ideal pipeline stages for four independent instructions

I DO — WORKED EXAMPLE

# Follow the worked method

**Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.**

### Explain the mechanism

2+3−1=4 cycles; serial baseline 2×3=6 cycles.

The second task starts stage one while first advances to stage two.





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**A pipeline makes each stage finish in less than one cycle by default.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

**Question 3**

Explain why branch and data hazards can reduce actual improvement, distinguishing latency and throughput. [6]



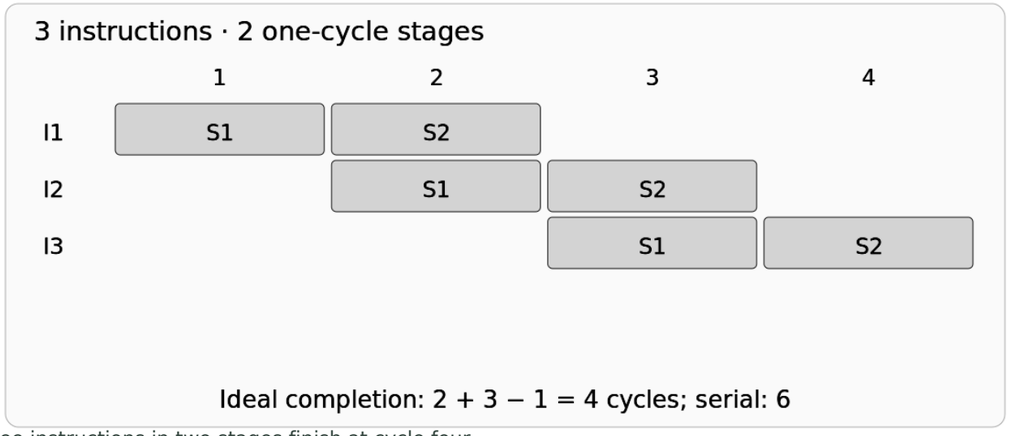


# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Three instructions pass through a two-stage pipeline. Compute ideal completion cycles.**

2+3−1=4 cycles; serial baseline 2×3=6 cycles.




Three instructions in two stages finish at cycle four

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Five independent instructions use a four-stage ideal pipeline. Calculate completion.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Show cycles for I1-I3 in a 3-stage F-D-E pipeline, and finish cycle of I3. [6]

### Question 2

Compute ideal pipeline versus serial completion for 6 instructions and 3 stages. [6]